MSE-2 CMPN-Solution

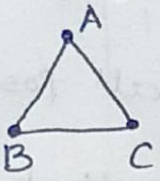
EM-III (BSC05)

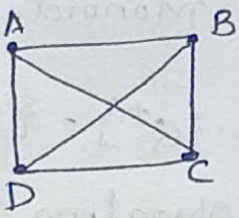
30 Marks / 1 hr.

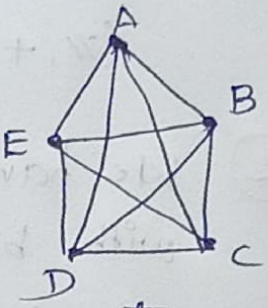
Q:1 Solve any Five (2 Marks Each)

(A) Regular Graph: A graph $G=G(V,E)$ in which every vertex has same degree is called as a regular graph.

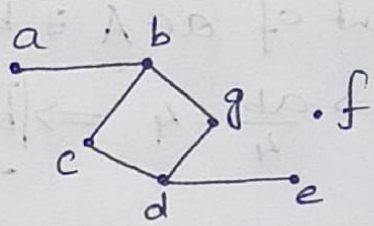
Eg:





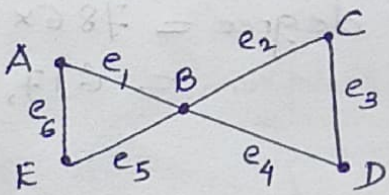


(B) A graph $G(V,E)$ in which the vertex with degree zero is called isolated vertex and a vertex with degree one is called as Pendant vertex.



Here vertices a & e are pendant vertices and vertex f is called isolated Vertex.

(C) Every Eulerian Graph is not Hamiltonian graph.



Graph $G(V,E)$ is Eulerian graph with Eulerian Circuit

$A\, e_1\, B\, e_2\, C\, e_3\, D\, e_4\, B\, e_5\, E\, e_6\, A$

But For Hamiltonian graph vertex B is repeated to travell all vertices.

So, $G(V,E)$ is not Hamiltonian graph.

(D) We have $\mathbb{Z}=\{\ldots,-3,-2,-1,0,1,2,3,\ldots\}$ is a Set of integers with binary operation '+'.

G_1: $\forall\, a,b \in \mathbb{Z}$, $a*b=a+b \in \mathbb{Z}$.

Hence, $(\mathbb{Z}, +)$ is groupoid.

G_2: $\forall a, b, c \in \mathbb{Z}$, $a+(b+c) = (a+b)+c$.

Hence, $(\mathbb{Z}, +)$ is a semi group.

G_3: $\forall a \in \mathbb{Z}$ $\exists$ $0 \in \mathbb{Z}$ such that $a+0 = 0+a = a$

Hence, '0' is called additive identity & $(\mathbb{Z}, +)$ is Monoid.

(E) We have $A = \{\mathbb{Q}^+\}$ = {set of all positive rational nos.} with binary operation $*$ such that $a*b = \frac{ab}{4}$, $\forall a, b \in A = \{\mathbb{Q}^+\}$.

(1) Let $\forall a \in A$, $\exists$ $e \in A$ such that $a*e = e*a = a$

$\frac{ae}{4} = a \Rightarrow e = 4 \in \mathbb{Q}^+$ is identity element.

(2) For Inverse element of $a \in A$ $\exists$ $b \in A$ such that

$a*b = e = 4 \Rightarrow \frac{ab}{4} = 4 \Rightarrow \boxed{b = \frac{16}{a}}$ is inverse of $a \in A$.

(F) A complete graph of vertices 786 has degree of each vertex is 785.

So, sum of all vertices degree $= \sum_{i=1}^{n} d(v_i) = 786 \times 785 = 617,010$.

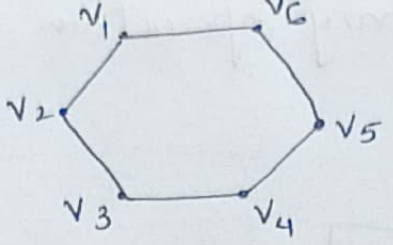
By Handshacking Lemma,

$\sum d(v_i) = 2$ [No. of Total Edges]

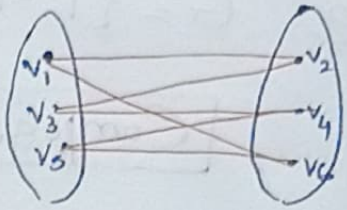
$617010 = 2 \times e$.

$e = 308,505$ = Total No. of Edges in complete graph.

(G)



YES → Bipartite Graph.

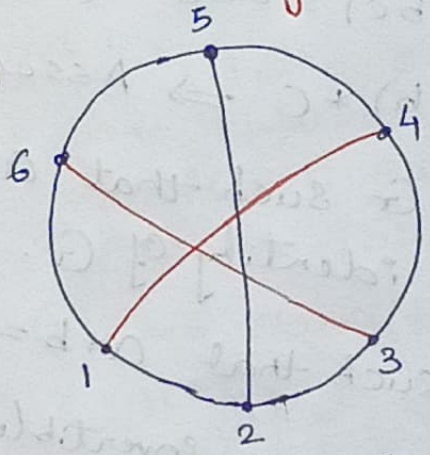


(H) Adjustment Matrix

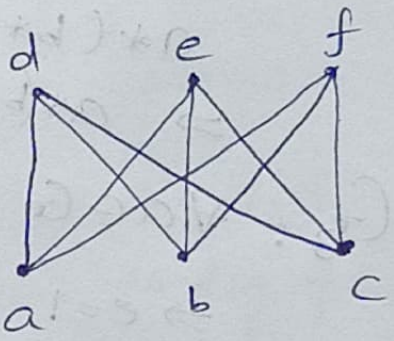
$$A_G = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Q:2: Solve Any Two (5 Marks Each).

(A)



Graph G



Graph H

For Isomorphic Graphs

	G	H	Remark
No. of vertices	6	6	Equal
No. of Edges	9	9	Equal
Deg of sequence	3, 3, 3, 3, 3, 3	3, 3, 3, 3, 3, 3	Equal
One to One bet^n vertices	$1 \to a$ $5 \to c$ $4 \to d$ $3 \to b$ $6 \to f$ $2 \to e$		validated
One to One bet^n Edges	$1 \to a$, $3 \to b$, $2 \to e$, $4 \to d$,	$5 \to c$ $6 \to f$	validated

∴ Graph G & Graph H are isomorphic.

(B) $G = \{1, -1, i, -i\}$ with binary operation $*$

Composite table

$*$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

G_1: $\forall a, b \in G$, $a*b = ab \in G \Rightarrow$ Closure prop exist

G_2: $\forall a, b, c \in G$ let $a=1, b=i$ & $c=-i$

$(a*b)*c = (ab)*c = abc = (1)(i)(-i) = 1$

$a*(b*c) = a*(bc) = abc = (1)(i)(-i) = 1$.

$\Rightarrow a*(b*c) = (a*b)*c \Rightarrow$ Associative exis

G_3: $\forall a \in G \ \exists \ e = 1 \in G$ such that $a*e = e*a =$

$\Rightarrow e = 1$ is mult identity of G.

G_4: $\forall a \in G \ \exists \ b \in G$ such that $a*b = ab = e = 1$

$a = 1/b \in G$ ($\because$ All are invertible)

Inverse exist.

Thus, $(G, *)$ is group under multiplication.

(C) Let $R = \{$set of Real Nos.$\}$ with Binary operation

$a*b = a+b+2$, $\forall a, b \in R$.

G_1: $\forall a, b \in R$, $a*b = a+b+2 \in R$

G_2: $\forall a, b, c \in R$, $(a*b)*c = (a+b+2)*c$

$= a+b+c+4$

& $a*(b*c) = a*(b+c+2) = a+b+c+4$

$\Rightarrow (a*b)*c = a*(b*c)$

G_3: $\forall a \in R \ \exists \ e \in R$ such that $a * e = a$

$\Rightarrow a + e + 2 = a \Rightarrow \boxed{e = -2} \in R$.

G_4: $\forall a \in R \ \exists \ b \in R$ such that $a * b = e$

$a + b + 2 = -2$.

$b = -4 - a \in R$.

Inverse exist.

G_5: $\forall a, b \in R$ $\quad a * b = a + b + 2$

$= b + a + 2$

$= b * a$

$\Rightarrow (R, *)$ is an Abelian Group.

Q.3 Solve any two [5 marks each].

(A) $G = \left\{ \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}, k \in \mathbb{Z} \right\}$ with usual mult^n $*$ of matrices.

Let $G = \{A, B, C, D, \ldots\}$.

$A = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$, $B = \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix}$, $C = \begin{bmatrix} c & 0 \\ 0 & c \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

G_1: $\forall A, B \in G$; $A * B = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}\begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} ab & 0 \\ 0 & ab \end{bmatrix} \in G$

G_2: $\forall A, B, C \in G$; $(A * B) * C = \begin{bmatrix} ab & 0 \\ 0 & ab \end{bmatrix}\begin{bmatrix} c & 0 \\ 0 & c \end{bmatrix} = \begin{bmatrix} abc & 0 \\ 0 & abc \end{bmatrix} \in G$

$A * (B * C) = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} * \begin{bmatrix} bc & 0 \\ 0 & bc \end{bmatrix} = \begin{bmatrix} abc & 0 \\ 0 & abc \end{bmatrix} \in G$

$\Rightarrow (A * B) * C = A * (B * C)$

G_3: $\forall A \in G \ \exists \ I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in G$ $\quad A * I = A$

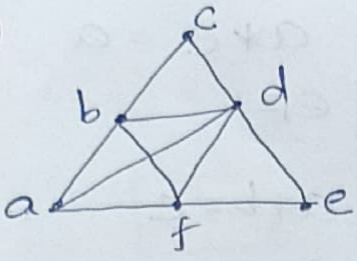
G_4: $\forall A \in G \ \exists \ B \in G$ such that $A * B = I$

$\Rightarrow \begin{bmatrix} ab & 0 \\ 0 & ab \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow ab = 1 \Rightarrow b = 1/a \notin \mathbb{Z}$

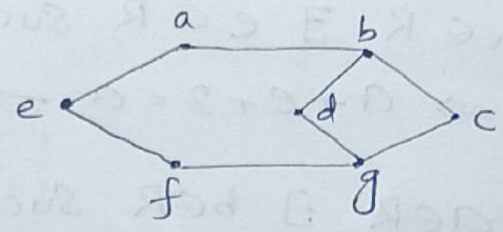
$\Rightarrow (G, *)$ is not a group.

But it is a Monoid.

3(B)



Graph G



Graph H

(6)

For graph G: ① Degree of vertices a & d are odd, Hence, graph G contain Eulerian path but no Eulerian Circuit. So, Graph G is not Eulerian Graph.

Eulerian Path: d-e-f-a-b-c-d-b-f-d-a

For graph H: deg. of vertex = 3 $\geq 6/2$

Graph has Hamiltonian Circuit so, it is a Hamiltonian Graph.

Hamiltonian Circuit: a-b-c-g-e-f-a

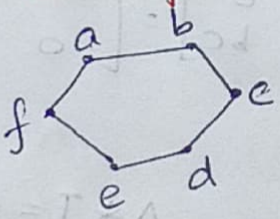
Graph H: ① deg of vertices b & g are odd hence it has Eulerian path but not Eulerian Circuit.

So, H is not Eulerian graph.

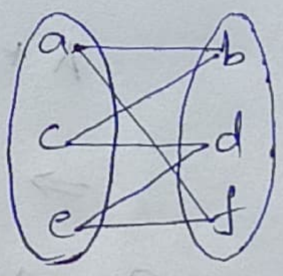
Eulerian Path: b-c-g-f-e-a-b-d-g

② No Hamiltonian Circuit as deg of each vertex $\not\geq \frac{n}{2}$.

3(C) ① Bipartite but not complete Bipartite graph.

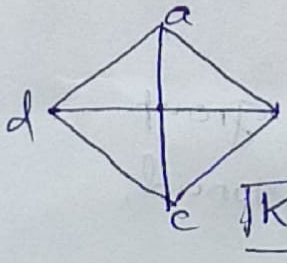


is Bipartite graph as



But not complete Bipartite graph as a is not connected with all members of other set.

② Hamiltonian But No Eulerian



K_4

③ Neither Eulerian nor Hamiltonian

